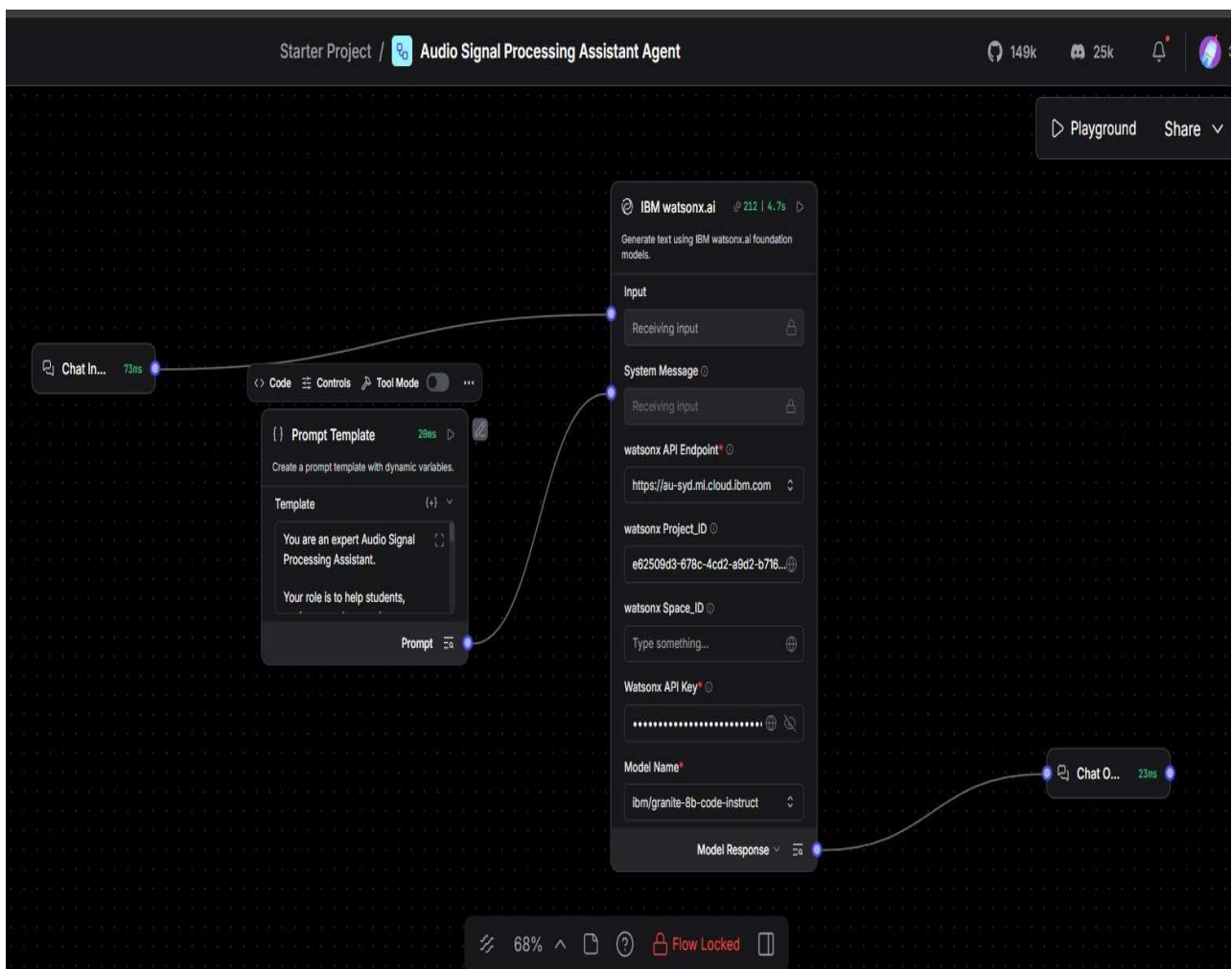


Audio Signal Processing Assistant Agent

Implementation Screenshots & Documentation

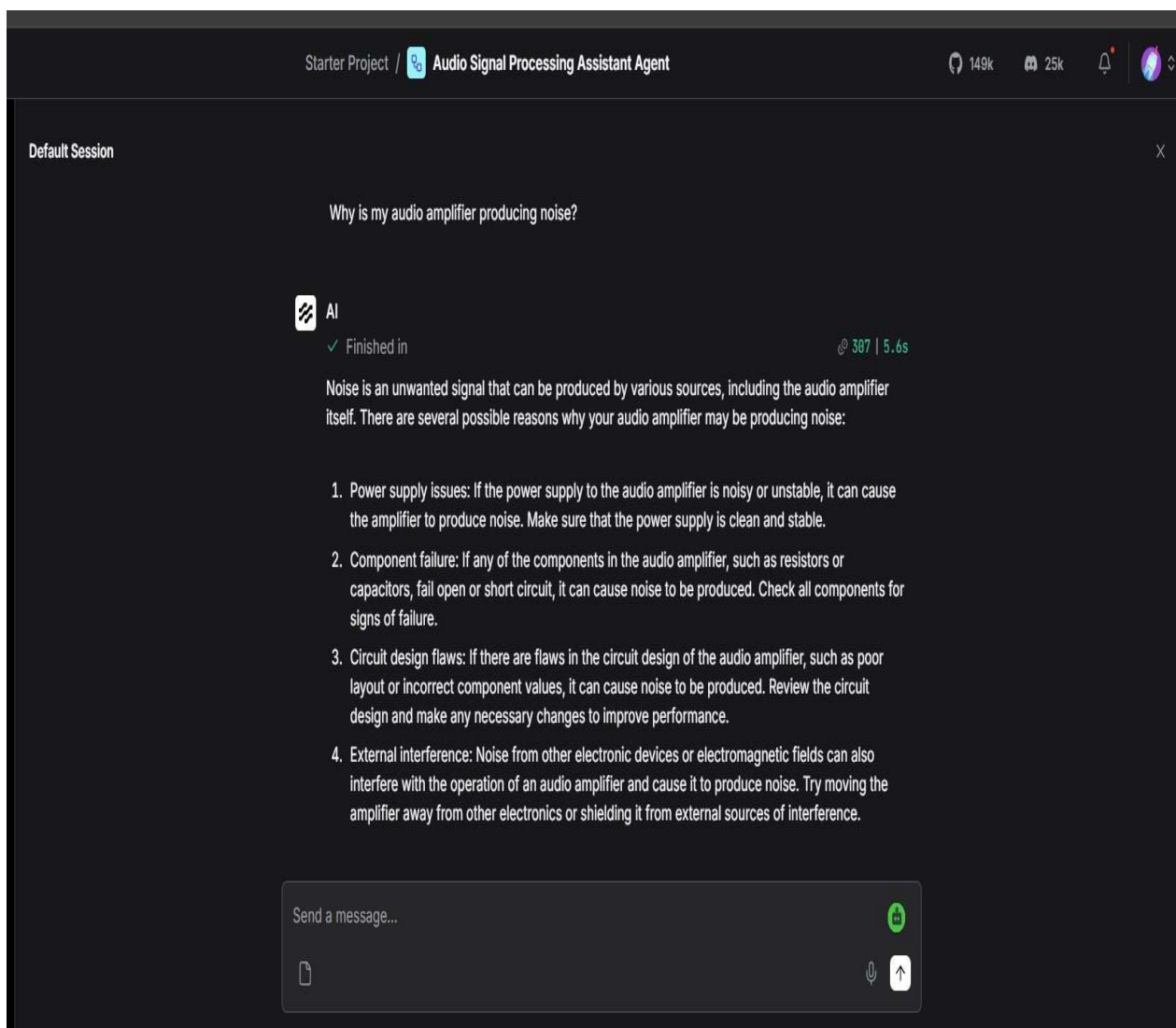
This document presents key screenshots from the development and testing of the Audio Signal Processing Assistant Agent, built using IBM WatsonX.ai, IBM Granite 8B, and LangFlow. Each figure is accompanied by a description of its purpose and significance.

Figure 1: LangFlow Workflow for Audio Signal Processing Assistant Agent



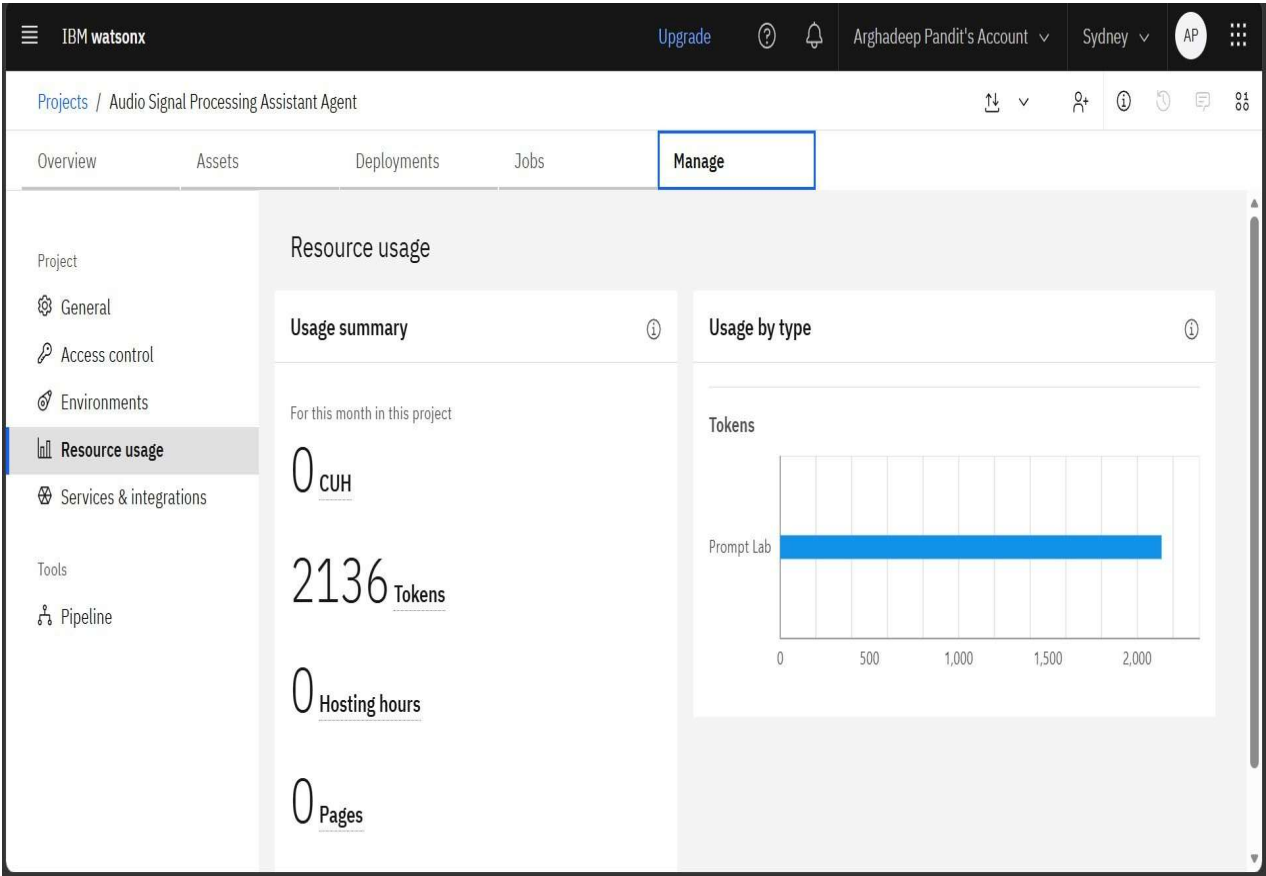
This figure illustrates the complete workflow of the Audio Signal Processing Assistant Agent developed using LangFlow. The architecture consists of a Chat Input component, a Prompt Template containing domain-specific instructions, IBM Watsonx.ai as the Large Language Model (LLM), and a Chat Output component. User queries are processed through the prompt engineering layer and analysed by the IBM Granite model to generate intelligent audio troubleshooting and signal-processing recommendations.

Figure 2: Agent Response Validation in LangFlow Playground



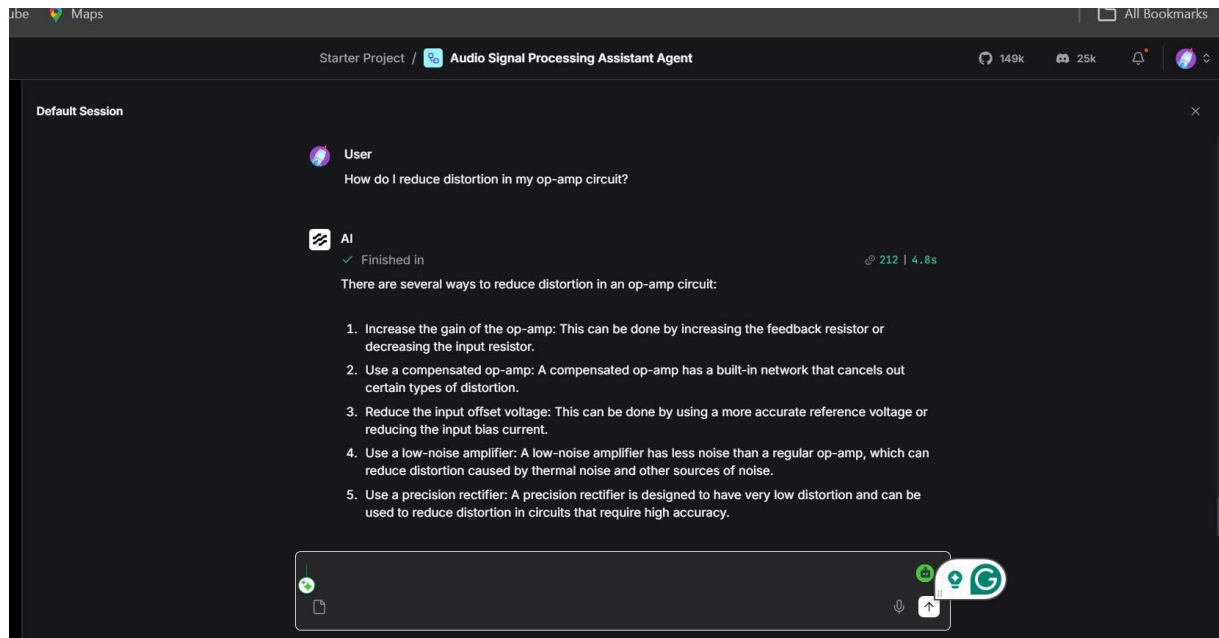
This figure demonstrates the successful execution and testing of the Audio Signal Processing Assistant Agent in the LangFlow Playground environment. A user query regarding audio amplifier noise was processed by the IBM Granite foundation model through IBM watsonx.ai, and the agent generated a detailed troubleshooting response.

Figure 3: IBM Watsonx.ai Project Dashboard — Resource Usage & Token Tracking



This figure demonstrates project monitoring, resource management, and token usage tracking within the IBM WatsonX.ai development environment. The dashboard displays the project's resource usage summary, including token consumption (2136 tokens) and usage breakdown by type, confirming active integration and successful API calls made during development and testing.

Figure 4: Agent Response — Op-Amp Distortion Troubleshooting



This figure demonstrates the agent's capability to analyze electronic circuit problems and provide intelligent troubleshooting recommendations for audio signal processing systems. The user query 'How do I reduce distortion in my op-amp circuit?' was answered with five structured solutions, showcasing the agent's domain-specific reasoning powered by the IBM Granite 8B model.